Chapter 4: IMU Pre-integration

Derivation of the Three Pre-integrated Increments

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1 Notation and assumptions

We derive the three pre-integrated increments corresponding to the textbook's equations (4.7), starting from the discrete-time IMU kinematics (4.4) and their accumulated form (4.6). Unlike the textbook's constant sampling interval Δt, we allow unequal sample intervals:

$$\boxed{\Delta t_k = t_{k+1} - t_k, \qquad \Delta t_{ij} = t_j - t_i = \sum_{k=i}^{j-1} \Delta t_k.} \tag{1}$$

We assume $R_k \in \mathrm{SO}(3)$ maps body-frame vectors to world-frame vectors, $p_k, v_k, g \in \mathbb{R}^3$ are world-frame quantities, gravity g is constant over the interval, and the bias/noise terms are represented by $b_{g,k}, b_{a,k}, \eta_{gd,k}, \eta_{ad,k}$. The IMU measurements are $\tilde{\omega}_k$ and $\tilde{a}_k$. We use a zero-order-hold approximation over each sample interval for the corrected acceleration and angular velocity. All products of rotations are *time ordered*.

For compactness, define

$$\alpha_k := \tilde{a}_k - b_{a,k} - \eta_{ad,k}, \qquad \Omega_k := \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k)\,. \tag{2}$$

The single-step discrete-time kinematics are

$$R_{k+1} = R_k \Omega_k, \tag{3a}$$
$$v_{k+1} = v_k + g\Delta t_k + R_k \alpha_k \Delta t_k, \tag{3b}$$
$$p_{k+1} = p_k + v_k \Delta t_k + \tfrac{1}{2} g \Delta t_k^2 + \tfrac{1}{2} R_k \alpha_k \Delta t_k^2. \tag{3c}$$

2 Rotation increment ΔR_{ij}

Repeated substitution into (3)a gives

$$\begin{aligned} R_{i+1} &= R_i \Omega_i, \\ R_{i+2} &= R_{i+1}\Omega_{i+1} = R_i \Omega_i \Omega_{i+1}, \\ R_{i+3} &= R_i \Omega_i \Omega_{i+1} \Omega_{i+2}, \\ R_j &= R_i \prod_{k=i}^{j-1} \Omega_k. \end{aligned}$$

Left-multiplying by R_i^T and using $R_i^T R_i = I_3$ yields

$$\boxed{\Delta R_{ij} := R_i^T R_j = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k)\,.} \tag{4}$$

The product is $\Omega_i \Omega_{i+1} \cdots \Omega_{j-1}$; its factors cannot generally be reordered. In particular, $\Delta R_{ii} = I_3$ and

$$\Delta R_{ik} = R_i^T R_k, \qquad R_k = R_i \Delta R_{ik}. \tag{5}$$

The right-hand side of (4) depends on IMU readings, biases, and noise, but not on the starting attitude R_i.

3 Velocity increment Δv_{ij}

Summing (3)b from $k = i$ to $j - 1$ gives a telescoping sum on the left:

$$\begin{aligned} v_j - v_i &= \sum_{k=i}^{j-1} (v_{k+1} - v_k) \\ &= \sum_{k=i}^{j-1} g\Delta t_k + \sum_{k=i}^{j-1} R_k \alpha_k \Delta t_k \\ &= g\Delta t_{ij} + \sum_{k=i}^{j-1} R_k \alpha_k \Delta t_k. \end{aligned} \tag{6}$$

Substitute $R_k = R_i \Delta R_{ik}$ from (5):

$$\begin{aligned} v_j - v_i - g\Delta t_{ij} &= \sum_{k=i}^{j-1} R_i \Delta R_{ik} \alpha_k \Delta t_k \\ &= R_i \sum_{k=i}^{j-1} \Delta R_{ik} \alpha_k \Delta t_k. \end{aligned} \tag{7}$$

Left-multiply by R_i^T:

$$\boxed{\begin{aligned} \Delta v_{ij} &:= R_i^T (v_j - v_i - g\Delta t_{ij}) \\ &= \sum_{k=i}^{j-1} \Delta R_{ik} (\tilde{a}_k - b_{a,k} - \eta_{ad,k}) \Delta t_k. \end{aligned}} \tag{8}$$

In particular, for any intermediate k,

$$\Delta v_{ik} = R_i^T (v_k - v_i - g\Delta t_{ik}) \quad \Longrightarrow \quad v_k = v_i + g\Delta t_{ik} + R_i \Delta v_{ik}. \tag{9}$$

Thus Δv_{ij} is *not* simply $v_j - v_i$: it removes gravity's contribution and expresses the result in the starting body frame.

4 Position increment Δp_{ij}

4.1 Accumulate the position equation

Summing (3)c gives

$$p_j - p_i = \sum_{k=i}^{j-1} v_k \Delta t_k + \tfrac{1}{2} g \sum_{k=i}^{j-1} \Delta t_k^2 + \tfrac{1}{2} \sum_{k=i}^{j-1} R_k \alpha_k \Delta t_k^2. \tag{10}$$

Unlike the velocity case, the first sum contains intermediate velocities v_k. We eliminate them using (9).

4.2 Substitute the intermediate velocity

$$\begin{aligned} \sum_{k=i}^{j-1} v_k \Delta t_k &= \sum_{k=i}^{j-1} (v_i + g \Delta t_{ik} + R_i \Delta v_{ik}) \Delta t_k \\ &= v_i \Delta t_{ij} + g \sum_{k=i}^{j-1} \Delta t_{ik} \Delta t_k + R_i \sum_{k=i}^{j-1} \Delta v_{ik} \Delta t_k. \end{aligned} \tag{11}$$

Insert this into (10):

$$\begin{aligned} p_j - p_i = v_i \Delta t_{ij} + g \left[\sum_{k=i}^{j-1} \Delta t_{ik} \Delta t_k + \tfrac{1}{2} \sum_{k=i}^{j-1} \Delta t_k^2 \right] \\ + R_i \sum_{k=i}^{j-1} \Delta v_{ik} \Delta t_k + \tfrac{1}{2} \sum_{k=i}^{j-1} R_k \alpha_k \Delta t_k^2. \end{aligned} \tag{12}$$

4.3 Combine gravity terms using a telescoping sum

Since $\Delta t_{i,k+1} = \Delta t_{ik} + \Delta t_k$,

$$\begin{aligned} \Delta t_{i,k+1}^2 - \Delta t_{ik}^2 &= (\Delta t_{ik} + \Delta t_k)^2 - \Delta t_{ik}^2 \\ &= 2 \Delta t_{ik} \Delta t_k + \Delta t_k^2. \end{aligned} \tag{13}$$

Therefore

$$\Delta t_{ik} \Delta t_k + \tfrac{1}{2} \Delta t_k^2 = \tfrac{1}{2} (\Delta t_{i,k+1}^2 - \Delta t_{ik}^2). \tag{14}$$

Sum over all intervals:

$$\begin{aligned} \sum_{k=i}^{j-1} \left(\Delta t_{ik} \Delta t_k + \tfrac{1}{2} \Delta t_k^2 \right) &= \tfrac{1}{2} \sum_{k=i}^{j-1} (\Delta t_{i,k+1}^2 - \Delta t_{ik}^2) \\ &= \tfrac{1}{2} (\Delta t_{ij}^2 - \Delta t_{ii}^2) \\ &= \boxed{\tfrac{1}{2} \Delta t_{ij}^2}, \qquad \Delta t_{ii} = 0. \end{aligned} \tag{15}$$

This identity holds even when the Δt_k are unequal.

4.4 Factor out the initial rotation

Using $R_k = R_i \Delta R_{ik}$ in (12), and combining the gravity terms, yields

$$\begin{aligned} p_j - p_i = v_i \Delta t_{ij} + \tfrac{1}{2} g \Delta t_{ij}^2 \\ + R_i \sum_{k=i}^{j-1} \left[\Delta v_{ik} \Delta t_k + \tfrac{1}{2} \Delta R_{ik} \alpha_k \Delta t_k^2 \right] . \end{aligned} \tag{16}$$

Move the initial-state terms to the left and multiply by R_i^T:

$$\boxed{\begin{aligned} \Delta p_{ij} &:= R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2 \right) \\ &= \sum_{k=i}^{j-1} \left[\Delta v_{ik} \Delta t_k + \tfrac{1}{2} \Delta R_{ik} (\tilde{a}_k - b_{a,k} - \eta_{ad,k}) \Delta t_k^2 \right] . \end{aligned}} \tag{17}$$

Here Δv_{ik} is the *pre-integrated* velocity increment from i to k, not the world-frame velocity v_k. Its use removes dependence on the initial velocity v_i and initial rotation R_i from the sum.

5 Summary of the three increments

Recall $\Delta t_{ij} = \sum_{k=i}^{j-1} \Delta t_k$. The pre-integrated quantities are

$$\boxed{\Delta R_{ij}} = R_i^T R_j = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k}) \Delta t_k) , \tag{18a}$$

$$\boxed{\Delta v_{ij}} = R_i^T (v_j - v_i - g \Delta t_{ij}) = \sum_{k=i}^{j-1} \Delta R_{ik} \alpha_k \Delta t_k , \tag{18b}$$

$$\boxed{\Delta p_{ij}} = R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2 \right) = \sum_{k=i}^{j-1} \left(\Delta v_{ik} \Delta t_k + \tfrac{1}{2} \Delta R_{ik} \alpha_k \Delta t_k^2 \right) . \tag{18c}$$

All three right-hand expressions depend on IMU measurements, biases, noise, and sample intervals, not on the absolute initial state (R_i, v_i, p_i). The bias dependence remains; later pre-integration bias corrections address changes in the bias estimates.

6 Interpretation and implementation notes

1. ΔR_{ij} is a relative rotation. Δv_{ij} and Δp_{ij} are gravity-compensated increments expressed in the frame at i; they are not simply $v_j - v_i$ and $p_j - p_i$.
2. Initialization at $k = i$ is $\Delta R_{ii} = I_3$, $\Delta v_{ii} = 0$, and $\Delta p_{ii} = 0$.
3. For unequal sample intervals, replace every single-step Δt by Δt_k, while the accumulated duration is Δt_{ij}.
4. These are the *noise-containing* relative increments of the textbook's definition. Later, the chapter separates noisy quantities from pre-integrated measurements denoted with a tilde and derives their covariance.

7 Pre-integration measurement model through the textbook's (4.18)

7.1 Fixed-bias assumption and noise convention

For the interval $[t_i, t_j]$, the pre-integration computation uses fixed linearization biases $b_{g,i}$ and $b_{a,i}$, rather than the varying $b_{g,k}, b_{a,k}$ of the initial kinematics. The bias estimate can later be corrected using bias Jacobians; that calculation is outside the scope of these notes. Let

$$u_k := \tilde{\omega}_k - b_{g,i}, \qquad \theta_k := u_k \Delta t_k, \qquad J_{r,k} := J_r(\theta_k). \tag{19}$$

Here J_r denotes the right Jacobian of SO(3). The discrete noise samples are $\eta_{gd,k}$ and $\eta_{ad,k}$. The equations below retain only first-order noise terms and neglect products of two noise perturbations.

7.2 Separating rotation measurement and noise

The right-Jacobian first-order identity is

$$\operatorname{Exp}(\theta + \epsilon) \simeq \operatorname{Exp}(\theta) \operatorname{Exp}(J_r(\theta)\epsilon). \tag{20}$$

Setting $\epsilon = -\eta_{gd,k}\Delta t_k$ gives

$$\operatorname{Exp}((u_k - \eta_{gd,k})\Delta t_k) \simeq \operatorname{Exp}(u_k \Delta t_k) \operatorname{Exp}(-J_{r,k}\eta_{gd,k}\Delta t_k). \tag{21}$$

Define the noise-free computed measurement

$$\boxed{\Delta\tilde{R}_{ij} := \prod_{k=i}^{j-1} \operatorname{Exp}(u_k \Delta t_k), \qquad \Delta\tilde{R}_{ii} = I_3.} \tag{22}$$

For any $B \in \mathrm{SO}(3)$, the adjoint identity is

$$\operatorname{Exp}(\xi) B = B \operatorname{Exp}(B^T \xi). \tag{23}$$

For example, the first two increments become

$$\begin{aligned} \Delta R_{i,i+2} &\simeq \Delta\tilde{R}_{i,i+1} \operatorname{Exp}(-J_{r,i}\eta_{gd,i}\Delta t_i) \Delta\tilde{R}_{i+1,i+2} \operatorname{Exp}(-J_{r,i+1}\eta_{gd,i+1}\Delta t_{i+1}) \\ &= \Delta\tilde{R}_{i,i+2} \operatorname{Exp}(-\Delta\tilde{R}^T_{i+1,i+2} J_{r,i}\eta_{gd,i}\Delta t_i) \operatorname{Exp}(-J_{r,i+1}\eta_{gd,i+1}\Delta t_{i+1}). \end{aligned} \tag{24}$$

In particular, the noise accompanying the interval $i+1$ is $\eta_{gd,i+1}$, not $\eta_{gd,i}$; this corrects the index typo in the textbook's displayed (4.10). Repeatedly moving all nominal rotations left gives

$$\Delta R_{ij} \simeq \Delta\tilde{R}_{ij} \prod_{k=i}^{j-1} \operatorname{Exp}\left(-\Delta\tilde{R}^T_{k+1,j} J_{r,k}\eta_{gd,k}\Delta t_k\right). \tag{25}$$

Combining the small noise exponentials to first order, define

$$\boxed{\delta\phi_{ij} := \sum_{k=i}^{j-1} \Delta\tilde{R}^T_{k+1,j} J_{r,k}\eta_{gd,k}\Delta t_k,} \tag{26}$$

where $\Delta\tilde{R}_{jj} = I_3$. Consequently,

$$\boxed{\Delta R_{ij} \simeq \Delta\tilde{R}_{ij} \operatorname{Exp}(-\delta\phi_{ij}).} \tag{27}$$

The adjoint rearrangement (23) is exact; the approximation enters when separating and combining the noise exponentials.

7.3 Velocity measurement and accumulated noise

The relative velocity increment, with fixed accelerometer bias, is

$$\Delta v_{ij} = \sum_{k=i}^{j-1} \Delta R_{ik}((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k. \tag{28}$$

Using $\Delta R_{ik} \simeq \Delta\tilde{R}_{ik}\,\mathrm{Exp}(-\delta\phi_{ik})$ and $\mathrm{Exp}(-\delta\phi_{ik}) \simeq I - \delta\phi_{ik}^{\wedge}$,

$$\begin{aligned}\Delta v_{ij} &\simeq \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(I - \delta\phi_{ik}^{\wedge})((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k \\ &\simeq \sum_{k=i}^{j-1} \left[\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i}) + \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^{\wedge}\delta\phi_{ik} - \Delta\tilde{R}_{ik}\eta_{ad,k}\right]\Delta t_k. \end{aligned}\tag{29}$$

The sign of the rotation-noise term is *positive* here, since $-\delta\phi^{\wedge} f = f^{\wedge}\delta\phi$. Define

$$\boxed{\Delta\tilde{v}_{ij} := \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k,} \tag{30}$$

and

$$\boxed{\delta v_{ij} := \sum_{k=i}^{j-1} \left[-\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^{\wedge}\delta\phi_{ik} + \Delta\tilde{R}_{ik}\eta_{ad,k}\right]\Delta t_k.} \tag{31}$$

Then

$$\boxed{\Delta v_{ij} \simeq \Delta\tilde{v}_{ij} - \delta v_{ij}.} \tag{32}$$

This sign convention agrees with the textbook's (4.14): the cross-product order determines the apparent sign.

7.4 Position measurement and accumulated noise

Start with the previously derived position increment:

$$\Delta p_{ij} = \sum_{k=i}^{j-1} \left[\Delta v_{ik}\Delta t_k + \tfrac{1}{2}\Delta R_{ik}((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k^2\right]. \tag{33}$$

Substitute $\Delta v_{ik} \simeq \Delta\tilde{v}_{ik} - \delta v_{ik}$ and $\Delta R_{ik} \simeq \Delta\tilde{R}_{ik}(I - \delta\phi_{ik}^{\wedge})$:

$$\begin{aligned}\Delta p_{ij} \simeq \sum_{k=i}^{j-1} \Big[&\Delta\tilde{v}_{ik}\Delta t_k + \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k^2 - \delta v_{ik}\Delta t_k \\ &+ \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^{\wedge}\delta\phi_{ik}\Delta t_k^2 - \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2\Big]. \end{aligned}\tag{34}$$

Define the computed pre-integrated position measurement

$$\boxed{\Delta\tilde{p}_{ij} := \sum_{k=i}^{j-1} \left[\Delta\tilde{v}_{ik}\Delta t_k + \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k^2\right],} \tag{35}$$

and its accumulated noise

$$\boxed{\begin{aligned}\delta p_{ij} := \sum_{k=i}^{j-1} \Big[&\delta v_{ik}\Delta t_k - \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^{\wedge}\delta\phi_{ik}\Delta t_k^2 \\ &+ \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2\Big].\end{aligned}} \tag{36}$$

Hence

$$\boxed{\Delta p_{ij} \simeq \Delta\tilde{p}_{ij} - \delta p_{ij}.} \tag{37}$$

8 Summary: pre-integration measurement model

Collecting the preceding definitions and the original relative-state increments yields the textbook's three measurement relations through (4.18):

$$\boxed{\Delta\tilde{R}_{ij}} \simeq R_i^T R_j \operatorname{Exp}(\delta\phi_{ij}), \tag{38a}$$

$$\boxed{\Delta\tilde{v}_{ij}} \simeq R_i^T(v_j - v_i - g\Delta t_{ij}) + \delta v_{ij}, \tag{38b}$$

$$\boxed{\Delta\tilde{p}_{ij}} \simeq R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) + \delta p_{ij}. \tag{38c}$$

The positive sign in the rotation measurement is consistent with $\Delta R_{ij} \simeq \Delta\tilde{R}_{ij}\operatorname{Exp}(-\delta\phi_{ij})$: right-multiply by $\operatorname{Exp}(\delta\phi_{ij})$ to isolate $\Delta\tilde{R}_{ij}$. Unlike δv_{ij} and δp_{ij}, rotation noise enters multiplicatively on SO(3).

The three tilde quantities are computed from IMU measurements and the fixed bias linearization point. The three delta noise quantities are random perturbations; their covariance propagation and bias-update Jacobians will be developed later, and are *not* included here.

9 Recursive propagation of pre-integration noise

9.1 Conventions and initialization

The accumulated noise is ordered as

$$\eta_{ik} := \begin{bmatrix}\delta\phi_{ik}\\ \delta v_{ik}\\ \delta p_{ik}\end{bmatrix} \in \mathbb{R}^9, \qquad \eta_{d,k} := \begin{bmatrix}\eta_{gd,k}\\ \eta_{ad,k}\end{bmatrix} \in \mathbb{R}^6. \tag{39}$$

We use the fixed biases $b_{g,i}, b_{a,i}$ from the preceding section, and the sample interval $\Delta t_k = t_{k+1} - t_k$. The only additional shorthand is

$$J_{r,k} := J_r((\tilde{\omega}_k - b_{g,i})\Delta t_k)\,. \tag{40}$$

The initial accumulated errors and covariance vanish:

$$\eta_{ii} = 0, \qquad \Sigma_{ii} = 0_{9\times 9}. \tag{41}$$

9.2 Rotation noise recursion

The accumulated rotation perturbation from the preceding section is

$$\delta\phi_{i,k+1} = \sum_{\ell=i}^{k} \Delta\tilde{R}_{\ell+1,k+1}^T J_{r,\ell}\eta_{gd,\ell}\Delta t_\ell. \tag{42}$$

Separate the final term and use $\Delta\tilde{R}_{\ell+1,k+1} = \Delta\tilde{R}_{\ell+1,k}\Delta\tilde{R}_{k,k+1}$ for $\ell < k$:

$$\delta\phi_{i,k+1} = \sum_{\ell=i}^{k-1} \Delta\tilde{R}_{k,k+1}^T\Delta\tilde{R}_{\ell+1,k}^T J_{r,\ell}\eta_{gd,\ell}\Delta t_\ell + J_{r,k}\eta_{gd,k}\Delta t_k \tag{43}$$

$$= \boxed{\Delta\tilde{R}_{k,k+1}^T\delta\phi_{ik} + J_{r,k}\eta_{gd,k}\Delta t_k}. \tag{44}$$

The transpose of $\Delta\tilde{R}_{k,k+1}$ is the adjoint transport of the previously accumulated rotation noise to the newest right-tangent coordinates.

9.3 Velocity noise recursion

Using the preceding definition of δv and splitting off the last IMU interval gives

$$\begin{aligned}\delta v_{i,k+1} &= \sum_{\ell=i}^{k-1}\Big[-\Delta\tilde{R}_{i\ell}(\tilde{a}_\ell - b_{a,i})^\wedge \delta\phi_{i\ell} + \Delta\tilde{R}_{i\ell}\eta_{ad,\ell}\Big]\Delta t_\ell \\ &\quad + \Big[-\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik} + \Delta\tilde{R}_{ik}\eta_{ad,k}\Big]\Delta t_k \qquad (45)\\ &= \delta v_{ik} - \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k \\ &\quad + \Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k. \qquad (46)\end{aligned}$$

The velocity error therefore couples to the accumulated rotation error.

9.4 Position noise recursion

Similarly, split the last interval from the position noise sum:

$$\delta p_{i,k+1} = \delta p_{ik} + \delta v_{ik}\Delta t_k - \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k^2 + \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2. \qquad (47)$$

Notice that the δv_{ik} term is the *previous* accumulated velocity noise, consistent with the discrete position integration rule.

9.5 Nine-dimensional matrix recursion

Collecting the preceding three relations gives

$$\boxed{\eta_{i,k+1} = A_k\eta_{ik} + B_k\eta_{d,k}}, \qquad (48)$$

where the 9×9 transition matrix is

$$\boxed{A_k = \begin{bmatrix} \Delta\tilde{R}_{k,k+1}^T & 0 & 0 \\ -\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge\Delta t_k & I_3 & 0 \\ -\tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge\Delta t_k^2 & I_3\Delta t_k & I_3 \end{bmatrix}} \qquad (49)$$

and the 9×6 noise-input matrix is

$$\boxed{B_k = \begin{bmatrix} J_{r,k}\Delta t_k & 0 \\ 0 & \Delta\tilde{R}_{ik}\Delta t_k \\ 0 & \tfrac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2 \end{bmatrix}}. \qquad (50)$$

Rows are ordered $(\delta\phi, \delta v, \delta p)$ and columns of B_k are ordered (η_{gd}, η_{ad}). These blocks are the coefficients of the preceding three recursions, not independent assumptions. We have used the interval index k consistently; the book sometimes labels the last interval $j-1$.

10 Covariance propagation of pre-integrated measurements

10.1 Discrete IMU noise covariance

Let $Q_{d,k} := \mathrm{Cov}(\eta_{d,k})$. If gyroscope and accelerometer discrete noise are independent,

$$Q_{d,k} = \begin{bmatrix} Q_{g,d,k} & 0 \\ 0 & Q_{a,d,k} \end{bmatrix}. \qquad (51)$$

With continuous-time white-noise covariance densities $Q_{g,c}$ and $Q_{a,c}$, the interval-average discrete noise convention used in the textbook gives

$$Q_{g,d,k} = \frac{Q_{g,c}}{\Delta t_k}, \qquad Q_{a,d,k} = \frac{Q_{a,c}}{\Delta t_k}. \tag{52}$$

Thus the covariance of the *noise increments* entering the rotation/velocity equations scales with Δt_k after multiplication by B_k. Do not confuse this $Q_{d,k}$ with the accumulated pre-integration covariance Σ_{ik}.

10.2 Derivation from the recursion

Define $\Sigma_{ik} = \mathrm{Cov}(\eta_{ik})$. Assume the new IMU noise $\eta_{d,k}$ is zero-mean and uncorrelated with the previously accumulated error η_{ik}; treat A_k, B_k as fixed at the current linearization point. Then

$$\Sigma_{i,k+1} = \mathrm{Cov}(A_k \eta_{ik} + B_k \eta_{d,k}) \tag{53}$$

$$= A_k \Sigma_{ik} A_k^T + B_k Q_{d,k} B_k^T, \tag{54}$$

because the two cross-covariance terms vanish. Therefore,

$$\boxed{\Sigma_{i,k+1} = A_k \Sigma_{ik} A_k^T + B_k Q_{d,k} B_k^T, \qquad \Sigma_{ii} = 0.} \tag{55}$$

10.3 Pre-integration propagation flow

Receive IMU sample $(\tilde{\omega}_k, \tilde{a}_k, \Delta t_k)$

$\downarrow$

Update $\Delta\tilde{R}_{i,k+1}, \Delta\tilde{v}_{i,k+1}, \Delta\tilde{p}_{i,k+1}$

$\downarrow$

Compute $A_k,\ B_k,\ Q_{d,k}$

$\downarrow$

$\Sigma_{i,k+1} = A_k \Sigma_{ik} A_k^T + B_k Q_{d,k} B_k^T$

$\downarrow$

Continue until endpoint j

At the endpoint, the computed measurements $(\Delta\tilde{R}_{ij}, \Delta\tilde{v}_{ij}, \Delta\tilde{p}_{ij})$ and their 9×9 covariance Σ_{ij} describe the pre-integrated IMU measurement and its uncertainty. This completes the measurement-noise and covariance propagation of the textbook's Section 4.2.2. The next section begins the bias-update model.

11 Bias Update: First-Order Measurement Correction

11.1 General first-order approximation of a rotation-valued function

Let $R : \mathbb{R}^3 \to \mathrm{SO}(3)$ be a differentiable rotation-valued function of a parameter $\phi \in \mathbb{R}^3$. For a sufficiently small parameter perturbation $\delta\phi$, define its *right-local rotation increment* $f(\delta\phi)$ by

$$\boxed{R(\phi + \delta\phi) = R(\phi)\,\mathrm{Exp}\big(f(\delta\phi)\big).} \tag{56}$$

This is an exact local parameterization of the relative rotation, not yet a first-order approximation. Multiplying on the left by $R(\phi)^T$ and applying the local logarithm gives

$$\boxed{f(\delta\phi) = \mathrm{Log}\big(R(\phi)^T R(\phi + \delta\phi)\big)\,.} \tag{57}$$

Because $f(0) = \mathrm{Log}(I) = 0$, its ordinary Euclidean first-order Taylor expansion at the zero perturbation is

$$\begin{aligned} f(\delta\phi) &= f(0) + \left.\frac{\partial f}{\partial \delta\phi}\right|_{\delta\phi=0} \delta\phi + O(\|\delta\phi\|^2) \\ &= J_{R,\phi}\,\delta\phi + O(\|\delta\phi\|^2), \end{aligned} \tag{58}$$

where the *right-trivialized parameter Jacobian* is defined by

$$\boxed{J_{R,\phi} := \left.\frac{\partial}{\partial \delta\phi} \mathrm{Log}\big(R(\phi)^T R(\phi+\delta\phi)\big)\right|_{\delta\phi=0}.} \tag{59}$$

Substituting into (56) yields the first-order rotation-valued approximation

$$\boxed{R(\phi+\delta\phi) \simeq R(\phi)\,\mathrm{Exp}(J_{R,\phi}\delta\phi).} \tag{60}$$

Here ϕ is a generic *input parameter*; it is not the pre-integration rotation-noise variable $\delta\phi_{ij}$.

11.2 Application to the pre-integrated rotation measurement

In the textbook's bias-update equation, set the generic input parameter to the initial gyroscope-bias estimate $b_{g,i}$. The pre-integrated rotation $\Delta\tilde{R}_{ij}$ is a function of this bias because its IMU increments contain $\tilde{\omega}_k - b_{g,i}$. Thus, (60) gives

$$\boxed{\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i}) \simeq \Delta\tilde{R}_{ij}(b_{g,i})\,\mathrm{Exp}\big(J_{\Delta R,b_g}\delta b_{g,i}\big)\,,} \tag{61}$$

where

$$J_{\Delta R,b_g} := \left.\frac{\partial}{\partial \delta b_{g,i}} \mathrm{Log}\Big(\Delta\tilde{R}_{ij}(b_{g,i})^T \Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i})\Big)\right|_{\delta b_{g,i}=0}. \tag{62}$$

The textbook abbreviates this 3×3 right-trivialized Jacobian as $\partial\Delta\tilde{R}_{ij}/\partial b_{g,i}$. It is *not* an ordinary entrywise derivative of a 3×3 rotation matrix.

11.3 Velocity and position bias corrections

Unlike the rotation measurement, $\Delta\tilde{v}_{ij}$ and $\Delta\tilde{p}_{ij}$ are vectors in $\mathbb{R}^3$, so ordinary first-order Taylor expansions apply. Let the fixed bias linearization point be $(b_{g,i}, b_{a,i})$ and its correction be $(\delta b_{g,i}, \delta b_{a,i})$. Then

$$\boxed{\begin{aligned} &\Delta\tilde{v}_{ij}(b_{g,i}+\delta b_{g,i}, b_{a,i}+\delta b_{a,i}) \\ &\quad\simeq \Delta\tilde{v}_{ij}(b_{g,i}, b_{a,i}) + \frac{\partial\Delta\tilde{v}_{ij}}{\partial b_{g,i}}\delta b_{g,i} + \frac{\partial\Delta\tilde{v}_{ij}}{\partial b_{a,i}}\delta b_{a,i}. \end{aligned}} \tag{63}$$

Likewise,

$$\boxed{\begin{aligned} &\Delta\tilde{p}_{ij}(b_{g,i}+\delta b_{g,i}, b_{a,i}+\delta b_{a,i}) \\ &\quad\simeq \Delta\tilde{p}_{ij}(b_{g,i}, b_{a,i}) + \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{g,i}}\delta b_{g,i} + \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{a,i}}\delta b_{a,i}. \end{aligned}} \tag{64}$$

Equations (61)–(64) are the three first-order bias-correction relationships of the textbook's (4.27). The rotation correction uses an exponential to remain on SO(3), whereas velocity and position corrections use ordinary vector addition. These expressions *define the form* of the correction; the computation of the five bias Jacobians is reserved for the subsequent derivation.

11.4 Gyroscope-bias Jacobian: product expansion

We now derive the rotation correction corresponding to the textbook's (4.28)–(4.29), using the same variable sampling intervals Δt_k as in the preceding sections. Throughout the pre-integration interval, $b_{g,i}$ is the fixed bias linearization point, and $\delta b_{g,i}$ is a small change to that *same* bias for every IMU sample.

Start from the pre-integrated rotation as a function of the bias:

$$\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i}) = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,i} - \delta b_{g,i})\Delta t_k)\,. \tag{65}$$

For each sample, let $\theta_k = (\tilde{\omega}_k - b_{g,i})\Delta t_k$, so that $J_{r,k} = J_r(\theta_k)$. The right-Jacobian increment formula from BCH yields

$$\mathrm{Exp}(\theta_k - \delta b_{g,i}\Delta t_k) \simeq \mathrm{Exp}(\theta_k)\,\mathrm{Exp}(-J_{r,k}\delta b_{g,i}\Delta t_k). \tag{66}$$

The Jacobian $J_{r,k}$ must be retained because θ_k is the nominal increment, not a perturbation that vanishes with $\delta b_{g,i}$.

For two consecutive intervals, write $A = \Delta\tilde{R}_{i,i+1}$, $B = \Delta\tilde{R}_{i+1,i+2}$, $\epsilon_i = -J_{r,i}\delta b_{g,i}\Delta t_i$ and $\epsilon_{i+1} = -J_{r,i+1}\delta b_{g,i}\Delta t_{i+1}$. Then

$$\begin{aligned}
\Delta\tilde{R}^{\mathrm{new}}_{i,i+2} &\simeq A\,\mathrm{Exp}(\epsilon_i)B\,\mathrm{Exp}(\epsilon_{i+1}) \\
&= AB\,\mathrm{Exp}(B^T\epsilon_i)\,\mathrm{Exp}(\epsilon_{i+1}) \\
&\simeq AB\,\mathrm{Exp}(B^T\epsilon_i + \epsilon_{i+1}).
\end{aligned} \tag{67}$$

The middle equality uses the exact SO(3) adjoint identity $\mathrm{Exp}(\epsilon)B = B\,\mathrm{Exp}(B^T\epsilon)$. In the last step, BCH would more generally introduce $J_r^{-1}(B^T\epsilon_i)$ multiplying ϵ_{i+1}. However, both arguments are first order in $\delta b_{g,i}$, so

$$J_r^{-1}(B^T\epsilon_i)\epsilon_{i+1} = \epsilon_{i+1} + O(\|\delta b_{g,i}\|^2). \tag{68}$$

Thus, no additional nontrivial Jacobian is required when *combining* the small bias corrections to first order.

Repeating the same adjoint transport across the remaining increments, with $\Delta\tilde{R}_{jj} = I_3$, gives

$$\boxed{\begin{aligned}
&\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i}) \\
&\quad\simeq \Delta\tilde{R}_{ij}(b_{g,i})\,\mathrm{Exp}\left[-\left(\sum_{k=i}^{j-1}\Delta\tilde{R}^T_{k+1,j}J_{r,k}\Delta t_k\right)\delta b_{g,i}\right].
\end{aligned}} \tag{69}$$

This is the first-order rotation bias correction in the textbook's (4.28).

Comparing (69) with the previously defined right-trivialized linearization $\Delta\tilde{R}_{ij}(b+\delta b) \simeq \Delta\tilde{R}_{ij}(b)\,\mathrm{Exp}(J^{ij}_{R,b_g}\delta b)$ gives

$$\boxed{J^{ij}_{R,b_g} := \frac{\partial\Delta\tilde{R}_{ij}}{\partial b_{g,i}} = -\sum_{k=i}^{j-1}\Delta\tilde{R}^T_{k+1,j}J_{r,k}\Delta t_k.} \tag{70}$$

The derivative notation above refers to the *right-trivialized* 3×3 Jacobian, not an elementwise derivative of a rotation matrix.

11.5 Recursive form of the gyroscope-bias Jacobian

To obtain the textbook's (4.30), separate the last term of (70):

$$\begin{aligned} J^{ij}_{R,b_g} &= -\sum_{k=i}^{j-2} \Delta\tilde{R}^T_{k+1,j} J_{r,k} \Delta t_k - \Delta\tilde{R}^T_{j,j} J_{r,j-1} \Delta t_{j-1} \\ &= -\sum_{k=i}^{j-2} \Delta\tilde{R}^T_{k+1,j} J_{r,k} \Delta t_k - J_{r,j-1} \Delta t_{j-1}. \end{aligned} \tag{71}$$

For $k \leq j-2$, the nominal relative rotations compose as

$$\Delta\tilde{R}_{k+1,j} = \Delta\tilde{R}_{k+1,j-1} \Delta\tilde{R}_{j-1,j}, \qquad \Delta\tilde{R}^T_{k+1,j} = \Delta\tilde{R}^T_{j-1,j} \Delta\tilde{R}^T_{k+1,j-1}. \tag{72}$$

Substituting and factoring out the final rotation gives

$$\begin{aligned} J^{ij}_{R,b_g} &= \Delta\tilde{R}^T_{j-1,j} \left[-\sum_{k=i}^{j-2} \Delta\tilde{R}^T_{k+1,j-1} J_{r,k} \Delta t_k \right] - J_{r,j-1} \Delta t_{j-1} \\ &= \boxed{\Delta\tilde{R}^T_{j-1,j} J^{i,j-1}_{R,b_g} - J_{r,j-1} \Delta t_{j-1}}. \end{aligned} \tag{73}$$

Equivalently, for an incoming IMU interval $[t_k, t_{k+1}]$,

$$\boxed{J^{i,k+1}_{R,b_g} = \Delta\tilde{R}^T_{k,k+1} J^{ik}_{R,b_g} - J_{r,k} \Delta t_k, \qquad J^{ii}_{R,b_g} = 0_{3\times 3}.} \tag{74}$$

The first term transports the previously accumulated Jacobian to the new rotation reference, and the second adds the current interval's bias sensitivity. In the textbook's (4.30), the last $J_{r,k}\Delta t$ should be read as the contribution of the *last* interval, $J_{r,j-1}\Delta t_{j-1}$, when the left-hand side ends at j.

12 Velocity and Position Bias Jacobians

We now complete the five bias Jacobians of the pre-integrated measurements, corresponding to the textbook's (4.33)–(4.34). All Jacobians are evaluated at the fixed bias linearization point $(b_{g,i}, b_{a,i})$. We retain the sampling interval Δt_k.

12.1 Velocity with respect to accelerometer bias

From the measurement definition,

$$\Delta\tilde{v}_{ij} = \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik} (\tilde{a}_k - b_{a,i}) \Delta t_k. \tag{75}$$

The rotation increment does not depend on accelerometer bias. Thus

$$\boxed{J^{ij}_{v,b_a} := \frac{\partial \Delta\tilde{v}_{ij}}{\partial b_{a,i}} = -\sum_{k=i}^{j-1} \Delta\tilde{R}_{ik} \Delta t_k.} \tag{76}$$

12.2 Velocity with respect to gyroscope bias

The gyro bias affects the velocity through the pre-integrated rotation. Apply the right-trivialized perturbation derived earlier:

$$\Delta\tilde{R}_{ik}(b_{g,i}+\delta b_g) \simeq \Delta\tilde{R}_{ik}\,\mathrm{Exp}(J^{ik}_{R,b_g}\delta b_g). \tag{77}$$

Keeping only first-order terms,

$$\begin{aligned}\Delta\tilde{v}_{ij}(b_{g,i}+\delta b_g) &\simeq \sum_{k=i}^{j-1}\Delta\tilde{R}_{ik}\left[I+(J^{ik}_{R,b_g}\delta b_g)^\wedge\right](\tilde{a}_k-b_{a,i})\Delta t_k \\ &= \Delta\tilde{v}_{ij}(b_{g,i}) - \sum_{k=i}^{j-1}\Delta\tilde{R}_{ik}(\tilde{a}_k-b_{a,i})^\wedge J^{ik}_{R,b_g}\delta b_g\Delta t_k.\end{aligned} \tag{78}$$

Here $u^\wedge w = -w^\wedge u$. Comparing with the first-order Taylor expansion gives

$$\boxed{J^{ij}_{v,b_g} := \frac{\partial\Delta\tilde{v}_{ij}}{\partial b_{g,i}} = -\sum_{k=i}^{j-1}\Delta\tilde{R}_{ik}(\tilde{a}_k-b_{a,i})^\wedge J^{ik}_{R,b_g}\Delta t_k.} \tag{79}$$

12.3 Position with respect to accelerometer bias

Recall

$$\Delta\tilde{p}_{ij} = \sum_{k=i}^{j-1}\left[\Delta\tilde{v}_{ik}\Delta t_k + \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k-b_{a,i})\Delta t_k^2\right]. \tag{80}$$

Differentiate both terms. The rotation has no dependence on $b_{a,i}$:

$$\boxed{J^{ij}_{p,b_a} := \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{a,i}} = \sum_{k=i}^{j-1}\left[J^{ik}_{v,b_a}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2\right].} \tag{81}$$

12.4 Position with respect to gyroscope bias

There are two dependency paths: gyro bias affects the accumulated velocity $\Delta\tilde{v}_{ik}$ and the rotation $\Delta\tilde{R}_{ik}$ in the acceleration contribution. Under $b_{g,i}\mapsto b_{g,i}+\delta b_g$, insert

$$\Delta\tilde{v}_{ik}(b_{g,i}+\delta b_g) \simeq \Delta\tilde{v}_{ik} + J^{ik}_{v,b_g}\delta b_g, \tag{82}$$

$$\Delta\tilde{R}_{ik}(b_{g,i}+\delta b_g) \simeq \Delta\tilde{R}_{ik}\left[I+(J^{ik}_{R,b_g}\delta b_g)^\wedge\right]. \tag{83}$$

After collecting the terms linear in δb_g,

$$\begin{aligned}\Delta\tilde{p}_{ij}(b_{g,i}+\delta b_g) &\simeq \Delta\tilde{p}_{ij}(b_{g,i}) \\ &\quad + \sum_{k=i}^{j-1}\left[J^{ik}_{v,b_g}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k-b_{a,i})^\wedge J^{ik}_{R,b_g}\Delta t_k^2\right]\delta b_g.\end{aligned} \tag{84}$$

Therefore,

$$\boxed{\begin{aligned}&J^{ij}_{p,b_g} := \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{g,i}} \\ &= \sum_{k=i}^{j-1}\left[J^{ik}_{v,b_g}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k-b_{a,i})^\wedge J^{ik}_{R,b_g}\Delta t_k^2\right].\end{aligned}} \tag{85}$$

13 Recursive Forms of the Five Bias Jacobians

For a new IMU interval $[t_k, t_{k+1}]$, split the last term from each sum. The rotation Jacobian has already been derived. The complete set of recursions is

$$J^{i,k+1}_{R,b_g} = \Delta\tilde{R}^T_{k,k+1} J^{ik}_{R,b_g} - J_{r,k}\Delta t_k, \tag{86}$$

$$J^{i,k+1}_{v,b_a} = J^{ik}_{v,b_a} - \Delta\tilde{R}_{ik}\Delta t_k, \tag{87}$$

$$J^{i,k+1}_{v,b_g} = J^{ik}_{v,b_g} - \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge J^{ik}_{R,b_g}\Delta t_k, \tag{88}$$

$$J^{i,k+1}_{p,b_a} = J^{ik}_{p,b_a} + J^{ik}_{v,b_a}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2, \tag{89}$$

$$J^{i,k+1}_{p,b_g} = J^{ik}_{p,b_g} + J^{ik}_{v,b_g}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge J^{ik}_{R,b_g}\Delta t_k^2. \tag{90}$$

All five Jacobians are initially zero at $k = i$. The right-hand sides use the *old* values at k, including $\Delta\tilde{R}_{ik}$ and the old velocity and rotation Jacobians. In an in-place implementation, update the position Jacobians first, then the velocity Jacobians, and finally the rotation Jacobian, or cache all old quantities.

13.1 Summary

The rotation bias Jacobian is a right-trivialized SO(3) derivative; the other four Jacobians are ordinary Euclidean derivatives. Equations (86)–(90) allow the five bias sensitivities to be accumulated alongside the pre-integrated measurements and covariance. These notes stop at the textbook's (4.34); graph-optimization residual Jacobians are deferred.

14 Pre-integration Factor in Graph Optimization

14.1 Purpose and keyframe states

The pre-integration measurement model derived above can now be used as an IMU constraint between keyframes i and j. At each keyframe, we use the state variables

$$x_k = (R_k, p_k, v_k, b_{a,k}, b_{g,k}). \tag{91}$$

The purpose of this section is to define the *IMU pre-integration factor*, not yet to assemble a complete multi-sensor optimization problem. The pre-integrated measurements are $\Delta\tilde{R}_{ij}$, $\Delta\tilde{v}_{ij}$ and $\Delta\tilde{p}_{ij}$, with covariance Σ_{ij}.

14.2 Residual definition

The measurement model gives a predicted relative motion from the keyframe states. The residual compares this prediction against the pre-integrated IMU measurement. Following the textbook's residual convention, define

$$r_{\Delta R_{ij}} = \mathrm{Log}\left(\Delta\tilde{R}^T_{ij} R_i^T R_j\right), \tag{92a}$$

$$r_{\Delta v_{ij}} = R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij}, \tag{92b}$$

$$r_{\Delta p_{ij}} = R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) - \Delta\tilde{p}_{ij}. \tag{92c}$$

The rotation residual uses the $SO(3)$ logarithm instead of subtracting rotation matrices. When the state-predicted relative motion agrees with the pre-integrated measurements, all three residuals vanish.

Stack the three components in a nine-dimensional vector:

$$\boxed{r_{ij}^{\mathrm{IMU}} = \begin{bmatrix} r_{\Delta R_{ij}} \\ r_{\Delta v_{ij}} \\ r_{\Delta p_{ij}} \end{bmatrix} \in \mathbb{R}^9.} \tag{93}$$

The pre-integration covariance derived previously supplies the weighting for a least-squares IMU factor, schematically

$$E_{ij}^{\mathrm{IMU}} = \left(r_{ij}^{\mathrm{IMU}}\right)^T \Sigma_{ij}^{-1} r_{ij}^{\mathrm{IMU}}. \tag{94}$$

This is the standard weighted least-squares interpretation of the textbook's residual and covariance; the section primarily defines the residuals and factor connectivity.

14.3 Dependence on the IMU biases

Although the residual expressions explicitly display only the rotation, position and velocity states, their pre-integrated measurement terms depend on the biases at the starting keyframe:

$$\Delta\tilde{R}_{ij} = \Delta\tilde{R}_{ij}(b_{g,i}), \tag{95}$$

$$\Delta\tilde{v}_{ij} = \Delta\tilde{v}_{ij}(b_{g,i}, b_{a,i}), \tag{96}$$

$$\Delta\tilde{p}_{ij} = \Delta\tilde{p}_{ij}(b_{g,i}, b_{a,i}). \tag{97}$$

The five bias Jacobians derived in the preceding section provide first-order corrections to these measurements when the optimization updates the biases, without reintegrating all IMU samples. The IMU pre-integration factor therefore connects $(R_i, p_i, v_i, b_{a,i}, b_{g,i})$ to (R_j, p_j, v_j). The bias random walks between the two keyframes are represented by separate bias factors, as illustrated in the textbook's Figure 4-1.

14.4 Scope: IMU constraints versus other sensor factors

At this stage, we have formulated *only* the IMU pre-integration constraint and the associated bias random-walk model. We have not yet introduced GNSS, wheel-encoder or other sensor observation factors into the graph. The IMU factor constrains relative motion, rather than providing an absolute position or attitude reference by itself.

Later, a full graph may combine IMU pre-integration, bias random walk, GNSS, odometry and prior factors. The precise graph construction and implementation are addressed in later sections of the chapter.

14.5 Summary

The conceptual progression is

$$\boxed{\text{IMU samples} \;\longrightarrow\; (\Delta\tilde{R}_{ij}, \Delta\tilde{v}_{ij}, \Delta\tilde{p}_{ij}, \Sigma_{ij}) \;\longrightarrow\; r_{ij}^{\mathrm{IMU}} \;\longrightarrow\; \text{IMU factor.}}$$

The next step is to compute the residual Jacobians with respect to the graph's optimization variables. Those derivations are not included here.

15 Jacobians of Preintegration Residuals

15.1 Rotation residual: right perturbations

We now linearize the rotation residual of the IMU factor with respect to its endpoint rotations. Following the same convention as the preceding sections, perturb each rotation on the right:

$$R_i' = R_i \operatorname{Exp}(\phi_i), \qquad R_j' = R_j \operatorname{Exp}(\phi_j), \tag{98}$$

where $\phi_i, \phi_j \in \mathbb{R}^3$ are small optimization increments, not the pre-integration rotation noise $\delta\phi_{ij}$. Recall the residual

$$r_{\Delta R_{ij}} = \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j\Big). \tag{99}$$

The pre-integrated measurement is held fixed when differentiating with respect to the endpoint rotations.

15.2 Derivative with respect to the initial rotation

Perturb R_i while holding R_j fixed:

$$\begin{aligned} r_{\Delta R_{ij}}(R_i \,\mathrm{Exp}(\phi_i)) &= \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T (R_i \,\mathrm{Exp}(\phi_i))^T R_j\Big) \\ &= \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T \,\mathrm{Exp}(-\phi_i) R_i^T R_j\Big). \end{aligned} \tag{100}$$

The SO(3) adjoint identity

$$\mathrm{Exp}(\xi)Q = Q\,\mathrm{Exp}(Q^T\xi), \qquad Q \in \mathrm{SO}(3), \tag{101}$$

with $Q = R_i^T R_j$, moves the small perturbation to the right:

$$\mathrm{Exp}(-\phi_i)R_i^T R_j = R_i^T R_j \,\mathrm{Exp}(-R_j^T R_i \phi_i), \tag{102}$$

$$r_{\Delta R_{ij}}(R_i \,\mathrm{Exp}(\phi_i)) = \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j \,\mathrm{Exp}(-R_j^T R_i \phi_i)\Big). \tag{103}$$

By (99), $\Delta\tilde{R}_{ij}^T R_i^T R_j = \mathrm{Exp}(r_{\Delta R_{ij}})$. Use the first-order right-BCH identity

$$\mathrm{Log}(\mathrm{Exp}(a)\,\mathrm{Exp}(\epsilon)) \simeq a + J_r^{-1}(a)\epsilon, \qquad \epsilon \text{ small}. \tag{104}$$

Consequently,

$$\begin{aligned} r_{\Delta R_{ij}}(R_i \,\mathrm{Exp}(\phi_i)) &\simeq \mathrm{Log}\big(\mathrm{Exp}(r_{\Delta R_{ij}})\,\mathrm{Exp}(-R_j^T R_i \phi_i)\big) \\ &\simeq r_{\Delta R_{ij}} - J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i \phi_i. \end{aligned} \tag{105}$$

Comparing with a first-order Taylor expansion yields

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_i} = -J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i.} \tag{106}$$

The minus sign comes from $(R_i \,\mathrm{Exp}(\phi_i))^T = \mathrm{Exp}(-\phi_i)R_i^T$; the $R_j^T R_i$ term comes from the adjoint transformation.

15.3 Derivative with respect to the final rotation

Perturb R_j while holding R_i fixed:

$$\begin{aligned} r_{\Delta R_{ij}}(R_j \,\mathrm{Exp}(\phi_j)) &= \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j \,\mathrm{Exp}(\phi_j)\Big) \\ &= \mathrm{Log}\big(\mathrm{Exp}(r_{\Delta R_{ij}})\,\mathrm{Exp}(\phi_j)\big) \\ &\simeq r_{\Delta R_{ij}} + J_r^{-1}(r_{\Delta R_{ij}})\phi_j. \end{aligned} \tag{107}$$

Here the perturbation is already at the far right, so no adjoint rearrangement is required. Therefore,

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_j} = J_r^{-1}(r_{\Delta R_{ij}}).} \tag{108}$$

15.4 Summary of the two rotation-residual Jacobians

$$\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_i} = -J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i, \tag{109}$$

$$\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_j} = J_r^{-1}(r_{\Delta R_{ij}}). \tag{110}$$

Both are 3×3 Jacobians with respect to right perturbations. These are the endpoint-rotation derivatives corresponding to the textbook's (4.37)–(4.38). The following sections extend these results to the bias, velocity, and position residual Jacobians.

16 Remaining IMU-Factor Residual Jacobians

16.1 Rotation residual with respect to gyroscope bias

Let $b_{g,i}^0$ be the bias at which the pre-integrated measurements were originally computed. At the current optimization iterate, write $b_{g,i} = b_{g,i}^0 + \delta b_g$. Introduce an additional infinitesimal perturbation ϵ_g solely to differentiate the residual. The previously computed right-trivialized pre-integration Jacobian is

$$J_{R,b_g} = \frac{\partial}{\partial \epsilon_g} \operatorname{Log}\Big(\Delta\tilde{R}_{ij}(b_{g,i}^0)^T \Delta\tilde{R}_{ij}(b_{g,i}^0 + \epsilon_g)\Big)\Big|_{\epsilon_g=0}. \tag{111}$$

The first-order bias-corrected rotation measurement is

$$\Delta\tilde{R}'_{ij} = \Delta\tilde{R}_{ij} \operatorname{Exp}(J_{R,b_g}\delta b_g). \tag{112}$$

For clarity, set $u = J_{R,b_g}\delta b_g$ and $J_{r,b} = J_r(u)$. By the right-Jacobian BCH identity,

$$\begin{aligned} \Delta\tilde{R}_{ij}(b_{g,i}^0 + \delta b_g + \epsilon_g) &\simeq \Delta\tilde{R}_{ij} \operatorname{Exp}(u + J_{R,b_g}\epsilon_g) \\ &\simeq \Delta\tilde{R}'_{ij} \operatorname{Exp}(J_{r,b}J_{R,b_g}\epsilon_g). \end{aligned} \tag{113}$$

Write $Q = R_i^T R_j$ and $r' = \operatorname{Log}((\Delta\tilde{R}'_{ij})^T Q)$. Substitution into the rotation residual gives

$$\begin{aligned} r(\epsilon_g) &\simeq \operatorname{Log}\Big(\operatorname{Exp}(-J_{r,b}J_{R,b_g}\epsilon_g)(\Delta\tilde{R}'_{ij})^T Q\Big) \\ &= \operatorname{Log}\big(\operatorname{Exp}(-J_{r,b}J_{R,b_g}\epsilon_g)\operatorname{Exp}(r')\big) \\ &= \operatorname{Log}\big(\operatorname{Exp}(r')\operatorname{Exp}(-\operatorname{Exp}(r')^T J_{r,b}J_{R,b_g}\epsilon_g)\big) \\ &\simeq r' - J_r^{-1}(r')\operatorname{Exp}(r')^T J_{r,b}J_{R,b_g}\epsilon_g. \end{aligned} \tag{114}$$

The third line is the exact SO(3) adjoint identity; the last line is the first-order right-BCH formula. Therefore,

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \epsilon_g} = -J_r^{-1}(r')\operatorname{Exp}(r')^T J_{r,b}J_{R,b_g}.} \tag{115}$$

The expression is the Jacobian of the *bias-corrected residual* at the current iterate. If $\delta b_g = 0$, then $J_{r,b} = I_3$.

16.2 Velocity residual Jacobians

Define the state-predicted relative velocity

$$q_v := R_i^T(v_j - v_i - g\Delta t_{ij}), \qquad r_{\Delta v_{ij}} = q_v - \Delta\tilde{v}_{ij}. \tag{116}$$

At fixed pre-integrated measurement and fixed other states,

$$\boxed{\frac{\partial r_{\Delta v_{ij}}}{\partial v_i} = -R_i^T, \qquad \frac{\partial r_{\Delta v_{ij}}}{\partial v_j} = R_i^T.} \tag{117}$$

For a right perturbation $R_i' = R_i \operatorname{Exp}(\phi_i)$,

$$\begin{aligned} r_{\Delta v_{ij}}(R_i') &= \operatorname{Exp}(-\phi_i) q_v - \Delta\tilde{v}_{ij} \\ &\simeq (I_3 - \phi_i^\wedge) q_v - \Delta\tilde{v}_{ij} \\ &= r_{\Delta v_{ij}} + q_v^\wedge \phi_i. \end{aligned} \tag{118}$$

Consequently,

$$\boxed{\frac{\partial r_{\Delta v_{ij}}}{\partial \phi_i} = q_v^\wedge.} \tag{119}$$

Because the measurement is subtracted, its two bias Jacobians enter with negative signs:

$$\frac{\partial r_{\Delta v_{ij}}}{\partial b_{g,i}} = -J^{ij}_{v,b_g}, \qquad \frac{\partial r_{\Delta v_{ij}}}{\partial b_{a,i}} = -J^{ij}_{v,b_a}. \tag{120}$$

16.3 Position residual Jacobians

Let

$$q_p := R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2 \right), \qquad r_{\Delta p_{ij}} = q_p - \Delta\tilde{p}_{ij}. \tag{121}$$

The Euclidean derivatives are

$$\frac{\partial r_{\Delta p_{ij}}}{\partial p_i} = -R_i^T, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial p_j} = R_i^T, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial v_i} = -R_i^T \Delta t_{ij}. \tag{122}$$

Under $R_i' = R_i \operatorname{Exp}(\phi_i)$, the same first-order expansion gives

$$\begin{aligned} r_{\Delta p_{ij}}(R_i') &= \operatorname{Exp}(-\phi_i) q_p - \Delta\tilde{p}_{ij} \\ &\simeq r_{\Delta p_{ij}} + q_p^\wedge \phi_i, \end{aligned} \tag{123}$$

so

$$\boxed{\frac{\partial r_{\Delta p_{ij}}}{\partial \phi_i} = q_p^\wedge.} \tag{124}$$

Again the bias dependence enters through the pre-integrated measurement:

$$\frac{\partial r_{\Delta p_{ij}}}{\partial b_{g,i}} = -J^{ij}_{p,b_g}, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial b_{a,i}} = -J^{ij}_{p,b_a}. \tag{125}$$

17 Complete IMU-Factor Jacobian Matrix

Use the residual ordering $(r_{\Delta R}, r_{\Delta v}, r_{\Delta p})$ and optimization perturbation ordering $(\phi_i, \delta p_i, \delta v_i, \epsilon_g, \epsilon_a, \phi_j, \delta p_j, \delta v_j)$. Each variable block has three components, so the assembled Jacobian has dimensions 9×24.
For a compact presentation, write

$$W := J_r^{-1}(r'), \qquad G := -W \operatorname{Exp}(r')^T J_{r,b} J_{R,b_g}. \tag{126}$$

The rotation blocks below use the bias-corrected residual r'; for the other two residuals the measurement bias Jacobians are those derived in the previous section. We display the matrix in three 3×24 block rows to avoid a page-width overflow.

$$
\begin{aligned}
J_{ij} &= \frac{\partial r_{ij}}{\partial \delta X} = \begin{bmatrix} J_R \\ J_v \\ J_p \end{bmatrix}, \\
J_R &= \begin{bmatrix} -W R_j^T R_i & 0 & 0 & G & 0 & W & 0 & 0 \end{bmatrix}, \\
J_v &= \begin{bmatrix} q_v^\wedge & 0 & -R_i^T & -J_{v,b_g}^{ij} & -J_{v,b_a}^{ij} & 0 & 0 & R_i^T \end{bmatrix}, \\
J_p &= \begin{bmatrix} q_p^\wedge & -R_i^T & -R_i^T \Delta t_{ij} & -J_{p,b_g}^{ij} & -J_{p,b_a}^{ij} & 0 & R_i^T & 0 \end{bmatrix}.
\end{aligned}
\tag{127}
$$

Here every displayed block is 3×3. The position and velocity rows do not directly depend on R_j in this residual convention. The Jacobian above is evaluated at the current bias-corrected pre-integration; the symbols J_{v,b_a}^{ij}, J_{v,b_a}^{ij}, J_{p,b_g}^{ij}, and J_{p,b_a}^{ij} refer to the stored first-order bias sensitivities.

17.1 Scope and next steps

The present notes stop at the textbook's position residual Jacobians (4.43). The assembled matrix is a convenient grouping of the individual Jacobian blocks, rather than a separate equation in the textbook. No additional sensor factors are included here.